3.2.1 Cast out the digit nine from a given number to check if it is a multiple of nine.

EXAMPLE

Is 46908 multiple of 9?

46908 = 4+6+0+8
= 18
= 1 + 8
= 9

46908 is a multiple of 9 or divisible by nine.

Let us Know

Any number or combination of digits in that number which add to 9 can be cast out from the given number. Then the sum of remaining digits of the number is divisible by 9 or multiple of 9.

In addition problem, we can check the sum by casting out nines.

EXAMPLE 1

Check the following numbers whether it is a multiple of 9 or not

24689 = 2 + 4 + 6 + 8 = 20 (It's not a multiple of 9.)

9108 = 0 (It's a multiple of 9.)

3165 = 1 + 5 = 6 (It's not a multiple of 9.)

EXAMPLE 2

Check the addition fact

3356 + 4729 = 8085

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8 + 4 = 21

12 = 21

1 + 2 = 2 + 1

3 = 3

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